

Summary

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- 2 Documentation
- 3 Introduction by Problems
 - Safety Properties of C Programs
 - Importance of Domain
- 4 Overview of formal techniques and formal methods
- 5 Modelling Language
- 6 A Simple Example : Management of Students and Teachers
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 - The factorial function
 - Communicating agents
- 11 Summary on Events

Current Section Summary

- Event B : <http://www.event-b.org/>
- Atelier B : <http://www.atelierb.eu/>
- RODIN Platform : <http://www.event-b.org/platform.html>
- EB2ALL Toolset : <http://eb2all.loria.fr>
- RIMEL project : <http://rimel.loria.fr>
- The Modelling Language Event-B and related topics as lectures notes, tutorials, models <https://mery54.github.io/teaching/mosos/>

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```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main() {
    int x, y;
    // Seed the random number generator with the current time
    srand(time(NULL));
    // Generate a random number between 1 and 100
    x = rand() % 100 + 1;
    // Perform some calculations
    y = x / (100 - x);
    printf("x=%d--and--y=%d\n", x, y);
    return 0;
}
```

```
[kernel] Parsing bug0.c (with preprocessing)
[rte:annot] annotating function main
[wp] Running WP plugin...
[kernel:annot:missing-spec] FRAMAC_SHARE/libc/stdlib.h:299: Warning:
  Neither code nor explicit exits and terminates for function rand
  generating default clauses. See --generated-spec-* options for n
[kernel:annot:missing-spec] FRAMAC_SHARE/libc/stdlib.h:302: Warning:
  Neither code nor explicit exits and terminates for function srand
  generating default clauses. See --generated-spec-* options for n
[kernel:annot:missing-spec] FRAMAC_SHARE/libc/time.h:126: Warning:
  Neither code nor explicit exits and terminates for function time
  generating default clauses. See --generated-spec-* options for n
[kernel:annot:missing-spec] FRAMAC_SHARE/libc/stdio.h:211: Warning:
  Neither code nor explicit exits and terminates for function prin
  generating default clauses. See --generated-spec-* options for n
[wp] 8 goals scheduled
[wp] [Timeout] typed_main_call_printf_va_1_requires (Alt-Ergo)
[wp] [Timeout] typed_main_assert_rte_division_by_zero (Alt-Ergo)
[wp] Proved goals:      6 / 8
    Qed:                3
    Alt-Ergo 2.6.0:      3 (17ms-20ms-27ms)
    Timeout:             2
```

Function main

```
// Heisenbug
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
//
int main() {
    int x, y, i=0;
    for (i = 0; i <= 1000000; i++) {
        // Seed the random number generator with the current time
        srand(time(NULL));
        // Generate a random number between 1 and 100
        x = rand() % 100 + 1;
        printf("Result: -x=-%d\n", x);
        // Perform some calculations
        y = x / (100 - x);
        printf("Result: -i=%d--and-y=%d\n", i, y);
    }
    return 0;
}
```

RTE with frama-c but a modification

```
// Heisenbug
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#include <limits.h>

int main() {
    int x, y, i=0;

    for (i = 0; i <= INT_MAX; i++) {
        // Seed the random number generator with the current time
        srand(time(NULL)+i);

        // Generate a random number between 1 and 100
        x = rand() % 100 + 1;

        printf(" Result: -x=-%d\n", x);
        // Perform some calculations
        y = x / (100 - x);

        printf(" Result: -i=%d -x=-%d - -and -y=%d\n", i, x, y);
    }
}
```

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Our aim is to analyze what is implicit and what is explicit in formal modelling...

- **Semantics in modelling :**

- ▶ Semantics expressed by a *theory* (e.g. Event-B) used to formalize hardware and/or software systems
- ▶ Same theory is used for wide variety of heterogeneous systems

- **Semantics in domain :**

- ▶ Environment within which system evolve : application domain/context
- ▶ Information provided by domain is often associated while in operation
- ▶ Either assumed and omitted while formalising systems or hardcoded in formal models
- ▶ Same context is used for wide variety of heterogeneous systems



A case study for studying these properties

Nose Gear Velocity



- Estimated ground velocity of the aircraft should be available only if it is within 3 km/hr of the true velocity at some moment within past 3 seconds

Characterization of a System (I)

- NG velocity system :
 - ▶ **Hardware :**
 - *Electro-mechanical sensor* : detects rotations
 - *Two 16-bit counters* : Rotation counter, Milliseconds counter
 - *Interrupt service routine* : updates rotation counter and stores current time.
 - ▶ **Software :**
 - *Real-time operating system* : invokes update function every 500 ms
 - *16-bit global variable* : for recording rotation counter update time
 - *An update function* : estimates ground velocity of the aircraft.
- Input data available to the system :
 - ▶ *time* : in milliseconds
 - ▶ *distance* : in inches
 - ▶ *rotation angle* : in degrees
- Specified system performs velocity estimations in *imperial* unit system
- **Note** : expressed functional requirement is in *SI* unit system (km/hr).

What are the main properties to consider for formalization ?

- Two different types of data :
 - ▶ counters with modulo semantics
 - ▶ non-negative values for time, distance, and velocity
- Two dimensions : *distance* and *time*
- Many units : distance (inches, kilometers, miles), time (milliseconds, hours), velocity (kph, mph)
- And interaction among components

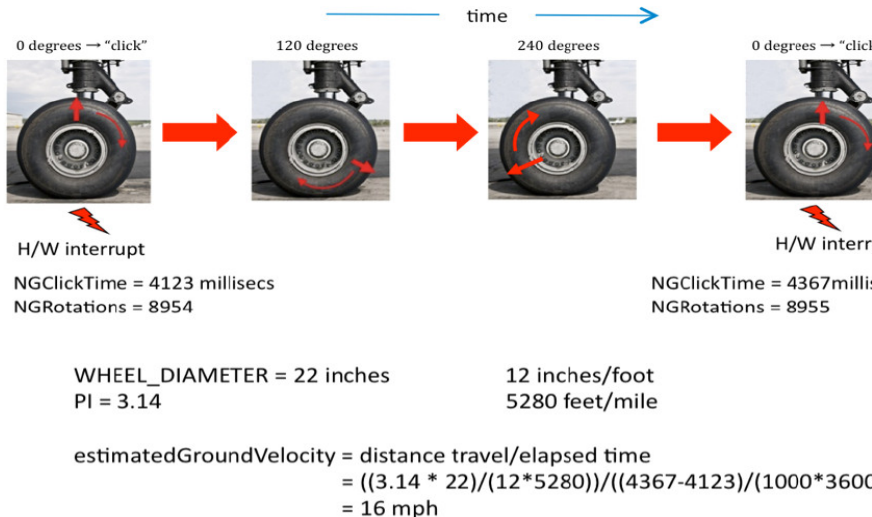
How should we model ?

- Designer needs to consider units and conversions between them to manipulate the model
- **One approach** : Model units as *sets*, and conversions as constructed types – *projections*.
- Example :

```
1 estimateVelocity ∈ MILES × HOURS → MPH
```

2 $mphTokph \in \text{MPH} \rightsquigarrow \text{KPH}$

Sample Velocity Estimation



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- Distributed systems : web services, information systems, distributed algorithms ...
- Safety critical systems : medical devices, embedded systems, cyber-physical systems, ...
- Fault-tolerant systems : networks, communication infrastructure, ...
- Environments : heart, the glucose-insulin regulatory system, ...

- Abstraction and refinement of features, 2000 *with D. Cansell*
- Incremental Proof of the Producer/Consumer Property for the PCI Protocol, 2002 *with D. Cansell, G. Gopalkrishnan, S. Jones.*
- A Mechanically Proved and Incremental Development of IEEE 1394 Tree Identify Protocol, 2003, *with J.-R. Abrial and D. Cansell.*
- The challenge of QoS for digital television services-. *EBU Technical Review* (avril 2005) *with D. Abraham, D. Cansell, C. Proch.*
- -Formal and Incremental Construction of Distributed Algorithms : On the Distributed Reference Counting Algorithm, 2006 *with D. Cansell.*

- Refinement : A Constructive Approach to Formal Software Design for a Secure e-voting Interface-, 2007 *with D. Cansell and P. Gibson.*
- Incremental Parametric Development of Greedy Algorithms, 2007, *with D. Cansell.*
- System-on-Chip Design by Proof-based Refinement, 2009 *with D. Cansell and C. Proch*
- -A simple refinement-based method for constructing algorithms, 2009. *Alone.*
- -Refinement-based guidelines for algorithmic systems-. *Alone. International Journal of Software and Informatics* (2009),

- Cryptologic algorithms : Event B development, combining cryptologic properties, modeling attacks.
- Access control systems : relating policy models and Event B models like in RBAC, TMAC, ORBAC
- Distributed algorithms : integration of local computation models into Event B, tool B2VISIDIA, algorithms of naming, election etc
- Medical devices : modelling the pacemaker, interacting with cardiologists, ...
- Modelling self- $\star$ systems
- Modelling medical devices item Modelling environments for medical devices : closed-loop modelling

Next modelling

- Modelling human-in-the-loop systems
- Modelling cyber-physical systems

General Approach

- Constructing a model of the system
- Elements for defining a formal or semi-formal model : syntax, semantics, verification, validation, documentation
- Mathematical structures : transition systems, temporal/modal/deontic/... logics,
- Validation of a model : tests, proofs, animation,...
- Modelling Techniques : state-based techniques
- Structure of a model : module, object, class,
- Design Patterns

Mathematical tools for modelling systems

- set theory : sets, relations, functions ...
- transition systems
- predicate calculus
- decision procedures
- interactive theorem prover

Examples of modelling languages

- Z : set theory, predicate calculus, schemas.
- VDM : types, pre/post specification, invariant, operations
- B : set theory, predicate calculus, generalized substitution, abstract machines, refinement, implementation.
- RAISE : abstract data types, functions,
- TLA⁺ : set theory, modules, temporal logic of actions.
- UNITY : temporal logic, actions systems, superposition.
- UML
- JML and Spec# : programming by contract

Objectives of the modelling

- To get a better understanding of the current system : requirements, properties, cost, maintenance . . .
- To document the the system
- To systematize operations of modelling : reuse, parametrization
- To ensure the quality of the final product : safety, security issues
- To elaborate a contract between the customer and the designer

The Triptych Approach

$$\mathcal{D}, \mathcal{S} \longrightarrow \mathcal{R} \quad (1)$$

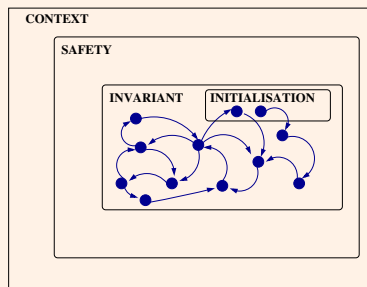
- $\mathcal{R}$ requirements or system properties
- $\mathcal{D}$ domain of the problem
- $\mathcal{S}$ model of the system
- $\longrightarrow$ relation of satisfaction

- Mathematical foundations of Models : syntax, semantics, pragmatics, theory, soundness.
- Mathematical reasoning is based on sound proof rules
- Common language for facilitating the communication.

Current Section Summary

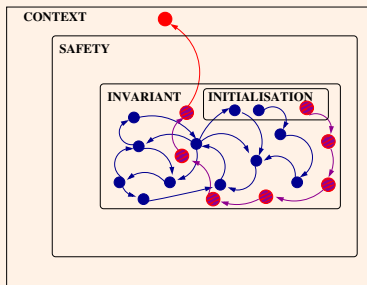
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Observing a safe system



- The context defines the possible values of states which are blue points.
- INVARIANT defines the set of effective possible values and corresponds to the visited or reachable states from the set INITIALISATION of initial states.
- Transitions modify state variables and maintains the invariant which mean that each reachable state is in the set INVARIANT of states.
- *Safety* requirement means that *something bad will never happen*.
- INVARIANT is a subset of SAFETY means that the system satisfies the safety property SAFETY.

Observing an unsafe system



- Transitions modify state variables and **may not maintain** the invariant.
- ... and **may not guarantee** safety properties.

- The red state is reachable and is not in the set SAFE.
- The system safety is defined by SAFE and transitions or events should be better controlled.

-

The Event B Method

- The Event B Method is invented by J.-R. Abrial from 1988 : abstract system, events, refinement, invariant.
- Atelier B and RODIN are supporting the Event B method
- An event is observed and triggered, when a guard is true
- Proof obligations are generated using the weakest-precondition semantics.
- A Event B model intends to model a reactive system.

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Managing teachers, students, lectures and class rooms

- Modelling the access control of students for lectures given by teachers
- When a student is attending a lecture, he/she can not attend another lecture
- When a teacher is lecturing, he/she is not lecturing another session.
- A student can not be lecturing without a teacher and when he is not attending a lecture, he is outside the classroom.
- When a teacher is ending a lecture, every student which is attending, is leaving the class room.
- When a student is not attending a lecture, he is free.

First step : identification of sets, constants, properties

- Sets : students, teachers
- **Property 1** : When a student is attending a lecture, he/she can not attend another lecture
- **Property 2** : When a teacher is lecturing, he/she is not lecturing another session.
- **Property 3** : A student can not be lecturing without a teacher and when he is not attending a lecture, he is outside the classroom.
- **Property 4** : When a teacher is ending a lecture, every student which is attending, is leaving the class room.
- **Property 5** : When a student is not attending a lecture, he is free.

Second step : definition of state variables

- The system model should be able to record the lecturing teachers and the attending students.
- The system model should be enough expressive to state when a given student is attending a lecture given by whom.
- Variable **attending** records students which attended some lecture with a given teacher.
- Variable **lecturing** records teachers who are lecturing.
- Variable **pause** records students are not attending a lecture but are somewhere not in a lecture.

Third step : properties of state variables

Expression of the invariant

$$x \triangleq (\textit{attending}, \textit{lecturing}, \textit{pause})$$

$$I(x) \triangleq \left\{ \begin{array}{l} \textit{inv1} : \textit{attending} \in \textit{STUDENTS} \leftrightarrow \textit{TEACHERS} \\ \textit{inv2} : \textit{lecturing} \subseteq \textit{TEACHERS} \\ \textit{inv3} : \forall e. e \in \textit{STUDENTS} \wedge e \in \textit{dom}(\textit{attending}) \\ \qquad \qquad \Rightarrow \textit{attending}(e) \in \textit{lecturing} \\ \textit{inv4} : \textit{pause} \subseteq \textit{STUDENTS} \\ \textit{inv5} : \textit{pause} \cap \textit{dom}(\textit{attending}) = \emptyset \\ \textit{inv6} : \textit{pause} \cup \textit{dom}(\textit{attending}) = \textit{STUDENTS} \end{array} \right.$$

Third step : properties of state variables

Expression of the invariant

$$x \triangleq (\textit{attending}, \textit{lecturing}, \textit{pause})$$

$$I(x) \triangleq \left\{ \begin{array}{l} \textit{inv1} : \textit{attending} \in \textit{STUDENTS} \mapsto \textit{TEACHERS} \\ \textit{inv2} : \textit{lecturing} \subseteq \textit{TEACHERS} \\ \textit{inv3} : \forall e \cdot e \in \textit{STUDENTS} \wedge e \in \textit{dom}(\textit{attending}) \\ \qquad \qquad \qquad \Rightarrow \textit{attending}(e) \in \textit{lecturing} \\ \textit{inv4} : \textit{pause} \subseteq \textit{STUDENTS} \\ \textit{inv5} : \textit{pause} \cap \textit{dom}(\textit{attending}) = \emptyset \\ \textit{inv6} : \textit{pause} \cup \textit{dom}(\textit{attending}) = \textit{STUDENTS} \end{array} \right.$$

Checking proof obligations

- For any event e , $I(x) \xrightarrow{e(x, x')} I(x')$
- For any event e , $I(x) \wedge e(x, x') \Rightarrow I(x')$

UseCases

- **EVENT** INITIALISATION : initializing state variables
- **EVENT** startingattending : a group of students is moving from *pause* to *lecture*
- **EVENT** teachergivinglecture : a teacher is starting a new lecture
- **EVENT** teacherendinglecture : a teacher is halting the lecture
- **EVENT** studentleavinglecture : a group of students is moving from *lecture* to *pause*

Fourth step : Updating state variables

EVENT INITIALISATION

BEGIN

$$act1 : attending := \emptyset$$
$$act2 : islecturing := \emptyset$$
$$act3 : pause := \overline{STUDENTS}$$

END

Fourth step : Updating state variables

```

EVENT startingattending
  ANY
     $e$   $e$  is a student
     $p$   $p$  is a teacher
  WHERE
     $grd1 : e \in STUDENTS$ 
     $grd3 : p \in TEACHERS$ 
     $grd4 : p \in islecturing$ 
     $grd2 : e \notin dom(attending)$ 
  THEN
     $act1 : attending(e) := p$ 
     $act2 : pause := pause \setminus \{e\}$ 
  END

```

Fourth step : Updating state variables

```

EVENT teachergivinglecture
  ANY
     $p$ 
  WHERE
     $grd2 : p \in TEACHERS$ 
     $grd1 : p \notin islecturing$ 
  THEN
     $act1 islecturing := islecturing \cup \{p\}$ 
END

```

Fourth step : Updating state variables

```

EVENT teacherendinglecture
  ANY
     $p$ 
  WHERE
     $grd1 : p \in TEACHERS$ 
     $grd2 : p \in islecturing$ 
  THEN
     $act1 : islecturing := islecturing \setminus \{p\}$ 
     $act3 : attending := attending \setminus \{f \mapsto q \mid \left( \begin{array}{l} f \in STUDENTS \\ \wedge q \in TEACHERS \\ \wedge f \mapsto q \in attending \\ \wedge q = p \end{array} \right)\}$ 
     $act2 : pause := pause \cup \{f \mid f \in STUDENTS \wedge f \in attending^{-1}[\{p\}]\}$ 
  END

```

Fourth step : Updating state variables

```
EVENT studentleavinglecture
ANY
  ge
WHERE
  grd1 :  $ge \subseteq \text{dom}(\text{attending})$ 
  grd2 :  $ge \neq \emptyset$ 
THEN
  act1 :  $\text{attending} := ge \triangleleft \text{attending}$ 
  act2 :  $\text{pause} := \text{pause} \cup ge$ 
END
```

The Rodin platform

The Rodin platform allows you to define a project called `ex0-school` and to define two components `school0` as context and `school1` as machine.

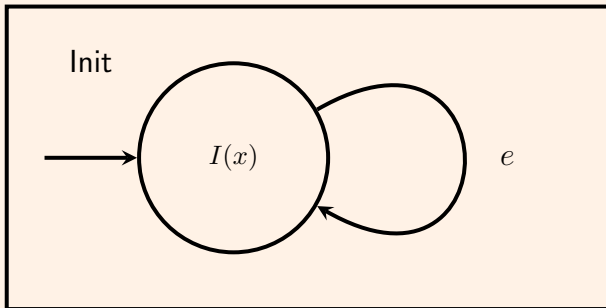
- sets and consthats are defined in the context.
- properties of sets and constants are defined as axioms or possibly theorems.
- Variables are declared in the machine and specified by the invariant
- Events are defined as possible changes of variables
- Observation may lead to add a new variable lectures which is typed as $lectures \in TEACHERS \rightarrow COURSES$

Mathematical tools for modelling systems

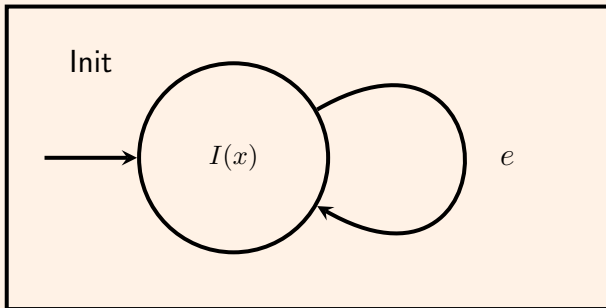
- set theory : sets, relations, functions ...
- transition systems
- predicate calculus
- decision procedures
- interactive theorem prover

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Summary on the internal checking

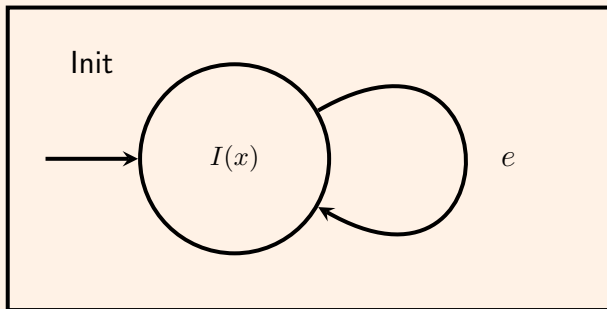


Summary on the internal checking



- Init satisfies the invariant $I(x)$ and updates the variable x with respect to some value satisfying Init or $Init(x) \Rightarrow I(x)$

Summary on the internal checking



- Init satisfies the invariant $I(x)$ and updates the variable x with respect to some value satisfying Init or $Init(x) \Rightarrow I(x)$
- e is an event observed and is interpreted as a relation among pre and post values and the transition means that :
$$I((x) \wedge BA(e)(x, x') \Rightarrow I(x')$$

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- A **system** is **observed**
- Observation of things which are
 - ▶ either changing over the **time** (*variable*)
 - ▶ or stuttering over the **time** (*constant*)
- A system is characterized by a **state**
- A state is made up of contextual **constant informations** over the problem theory and of **modifiable flexible informations** over the system.

Changing state of system

A **flexible variable** x is observed at different instants :

$$x_0 \xrightarrow{\tau} x_1 \xrightarrow{\tau} x_2 \xrightarrow{\tau} x_3 \xrightarrow{\tau} \dots \xrightarrow{\tau} x_i \xrightarrow{\tau} x_{i+1} \xrightarrow{\tau} \dots$$

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τ hides effective changes of state or actions or events

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$$x_0 \xrightarrow{\alpha_1} x_1 \xrightarrow{\alpha_2} x_2 \xrightarrow{\alpha_3} x_3 \xrightarrow{\alpha_4} \dots \xrightarrow{\alpha_i} x_i \xrightarrow{\alpha_{i+1}} x_{i+1} \xrightarrow{\alpha_{i+2}} \dots$$

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Occurrences of e τ can be added between two instants ie **stuttering steps** :

Changing state of system

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$$x_0 \xrightarrow{\tau} x_1 \xrightarrow{\tau} x_2 \xrightarrow{\tau} x_3 \xrightarrow{\tau} \dots \xrightarrow{\tau} x_i \xrightarrow{\tau} x_{i+1} \xrightarrow{\tau} \dots$$

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Occurrences of e τ can be added between two instants ie **stuttering steps** :

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A **safety** property S over x states that something will not happen : $S(x)$ means that S holds for x

An **invariant** property I over x states a strong safety property

$$\begin{aligned} x_0 &\xrightarrow{\alpha_1} x_1 \xrightarrow{\alpha_2} x_2 \xrightarrow{\tau} x_2 \xrightarrow{\alpha_3} x_3 \xrightarrow{\alpha_4} \dots \xrightarrow{\alpha_i} x_i \xrightarrow{\tau} x_i \xrightarrow{\alpha_{i+1}} x_{i+1} \xrightarrow{\alpha_{i+2}} \dots \\ (S(x_0) &\xrightarrow{\alpha_1} S(x_1) \xrightarrow{\alpha_2} S(x_2) \xrightarrow{\tau} S(x_2) \xrightarrow{\alpha_3} S(x_3) \xrightarrow{\alpha_4} \dots \xrightarrow{\alpha_i} S(x_i) \xrightarrow{\tau} \\ S(x_i) &\xrightarrow{\alpha_{i+1}} S(x_{i+1}) \xrightarrow{\alpha_{i+2}} \dots \end{aligned}$$

or equivalently $\forall i \in \mathbb{N} : S(x_i)$

Checking the relation

- You can check for every i in $\mathbb{N}$ that $S(x_i)$ is true but it can be long if states are different
- You can compute an abstraction of the set of states
- You can try to prove and for instance the induction principle may be usefull
- So be carefull and improve your modelling before to run the checker
- Use the induction

State properties of a system

- A state property namely $P(x)$ is a first order predicate with free variables x , where x is a flexible variable.
- A flexible variable x has a current value x , a next value x' , an initial value x_0 and possibly a final value x_f .
- A predicate $P(x)$ is considered as a set of values v such that $P(v)$ holds : set-theoretical interpretation

Safety Property

A safety property states that nothing bad can happen.

Example

Safety Properties

- Partial correctness a component is correct with respect to a precondition and a postcondition.
- No Run Time Error any software action or event does not produce a run-time error as overflow, division by zero ...
- Mutual exclusion a set of processes share common resources, a printer is shared by users, ...
- Deadlock freedom the system is never blocked, there is always at least one next state, ...

- An action α over states is a relation between values of state variables **before** and values of variables **after**

$$\alpha(x, x') \text{ or } x \xrightarrow{\alpha} x'$$

- Flexible variable x has two values x and x' .
- Priming flexible variables is borrowed from TLA
- **Hypothesis 1** : **Values of x belongs to a set of values called VALUES and defines the context of the system.**
- **Hypothesis 2** : **Relations over x and x' belong to a set of relations $\{r_0, \dots, r_n\}$**

Operational model of a system

- A system $\mathcal{S}$ is observed with respect to flexible variables x .
- Flexible variables x of $\mathcal{S}$ are modified according to a finite set of relations over the set of values $\text{VALUES} : \{r_0, \dots, r_n\}$
- $\text{INIT}(x)$ denotes the set of possible initial values for x .

$$\mathcal{OMS} = (x, \mathbf{Values}, \mathbf{Init}(x), \{r_0, \dots, r_n\})$$

- **Hypothesis 3** : $\mathcal{O}\mathcal{M}\mathcal{S} = (x, \text{VALUES}, \text{INIT}(x), \{r_0, \dots, r_n\})$
- **Hypothesis 4** : $x \longrightarrow x' \triangleq (x \ r_0 \ x') \vee \dots \vee (x \ r_n \ x')$
- $I(x)$ is inductively invariant for a system called $\mathcal{S}$, if

$$\begin{cases} \forall x \in \text{VALUES} : \text{INIT}(x) \Rightarrow I(x) \\ \forall x, x' \in \text{VALUES} : I(x) \wedge x \longrightarrow x' \Rightarrow I(x') \end{cases}$$
 $I(x)$ is called an invariant in Event B
- $Q(x)$ is a safety property for a system called $\mathcal{S}$, if

$$\forall x_0, x \in \text{VALUES} : \text{INIT}(x_0) \wedge x_0 \xrightarrow{*} x \Rightarrow Q(x)$$
 $Q(x)$ is called a theorem in Event B

Modelling systems : first attempt

MODEL

m

...

...

...

VARIABLES

x

INVARIANT

$I(x)$

THEOREMS

$Q(x)$

INITIALISATION

$Init(x)$

EVENTS

$\{r_0, \dots, r_n\}$

END

- A model has a name m
- Flexible variables x are declared
- $I(x)$ provides information over x
- $Q(x)$ provides information over x

- $\forall x_0, x \in \text{VALUES} : \text{INIT}(x_0) \wedge x_0 \xrightarrow{*} x \Rightarrow Q(x)$
- **Solution 1** Writing a procedure checking $\text{INIT}(x_0) \wedge x_0 \xrightarrow{*} x \Rightarrow Q(x)$ for each pair $x_0, x \in \text{VALUES}$, when VALUES is finite and small.
- **Solution 2** Writing a procedure checking $\text{INIT}(x_0) \wedge x_0 \xrightarrow{*} x \Rightarrow Q(x)$ for each pair $x_0, x \in \text{VALUES}$, by constructing an abstraction of VALUES .
- **Solution 3** Writing a proof for $\forall x_0, x \in \text{VALUES} : \text{INIT}(x_0) \wedge x_0 \xrightarrow{*} x \Rightarrow Q(x)$.

Defining an induction principle for an operational model

$$(I) \forall x_0, x \in \mathbf{Values} : \mathbf{Init}(x_0) \wedge x_0 \xrightarrow{\star} x \Rightarrow \mathbf{Q}(x)$$

if, and only if,

(II) there exists a state property $I(x)$ such that :

$$\forall x_0, x, x' \in \mathbf{Values} : \begin{cases} (1) \mathbf{Init}(x_0) \Rightarrow I(x_0) \\ (2) I(x) \Rightarrow \mathbf{Q}(x) \\ (3) I(x) \wedge x \longrightarrow x' \Rightarrow I(x') \end{cases}$$

if, and only if,

(III) there exists a state property $I(x)$ such that :

$$\forall x_0, x, x' \in \mathbf{Values} : \begin{cases} (1) \mathbf{Init}(x_0) \Rightarrow I(x_0) \\ (2) I(x) \Rightarrow \mathbf{Q}(x) \\ (3) \forall i \in \{0, \dots, n\} : I(x) \wedge x \ r_i \ x' \Rightarrow I(x') \end{cases}$$

Modelling systems : second attempt

MODEL m

...

...

...

VARIABLES x **INVARIANT** $I(x)$ **THEOREMS** $Q(x)$ **INITIALISATION** $Init(x)$ **EVENTS** $\{r_0, \dots, r_n\}$ **END**

- $\forall x_0 \in \text{VALUES} : \text{INIT}(x_0) \Rightarrow I(x_0)$
- $\forall x, x' \in \text{VALUES} : \forall i \in \{0, \dots, n\} :$
 $I(x) \wedge x \text{ } r_i \text{ } x' \Rightarrow I(x')$
- $\forall x \in \text{VALUES} : I(x) \Rightarrow Q(x)$

Modelling systems : last attempt ?

MODEL

 m

?

?

?

VARIABLES

 x

INVARIANT

$$I(x)$$

THEOREMS

$$Q(x)$$

INITIALISATION

$$Init(x)$$

EVENTS

$$\{r_0, \dots, r_n\}$$

END

- What are the environment of the proof for properties?
- What are theories?
- How are defining the static objects?

Modelling systems : last attempt !

MODEL

 m
$$\Gamma(m)$$

VARIABLES

 x

INVARIANT

$$I(x)$$

THEOREMS

$$Q(x)$$

INITIALISATION

$$Init(x)$$

EVENTS

$$\{r_0, \dots, r_n\}$$

END

- $\Gamma(m)$ defines the static environment for the proofs related to m .
- $\Gamma(m) \vdash \forall x \in \text{VALUES} : \text{INIT}(x) \Rightarrow \text{I}(x)$
- $\forall i \in \{0, \dots, n\} :$
 $\Gamma(m) \vdash \forall x, x' \in \text{VALUES} : \text{I}(x) \wedge x \text{ } r_i \text{ } x' \Rightarrow \text{I}(x')$
- $\Gamma(m) \vdash \forall x \in \text{VALUES} : \text{I}(x) \Rightarrow \text{Q}(x)$

An **event system model** is made of

State **constants** and state **variables** constrained by a state **invariant**

A finite set of **events**

Proofs ensures the consistency between the invariant and the events

An event system model can be **refined**

Proofs must ensure the correctness of refinement

Modelling systems : Hello world !

MODEL FACTORIAL_EVENTS

Static Part *context*

CONSTANTS

factorial, m

AXIOMS

$$\begin{aligned} & m \in \mathbb{N} \wedge factorial \in \mathbb{N} \leftrightarrow \mathbb{N} \wedge 0 \mapsto 1 \in factorial \wedge \\ & \forall(n, fn).(n \mapsto fn \in factorial \Rightarrow n + 1 \mapsto (n + 1) * fn \in factorial) \wedge \\ & \forall f. \left(\begin{array}{l} f \in \mathbb{N} \leftrightarrow \mathbb{N} \wedge \\ 0 \mapsto 1 \in f \wedge \\ \forall(n, fn).(n \mapsto fn \in f \Rightarrow n + 1 \mapsto (n + 1) \times fn \in f) \\ \Rightarrow \\ factorial \subseteq f \end{array} \right) \end{aligned}$$

Dynamic Part *machine*

VARIABLES

result, ok

INVARIANT

result $\in \mathbb{N}$

ok $\in \mathbb{B}$

ok = TRUE $\Rightarrow$ *result* = *factorial*(*n*)

THEOREMS

factorial $\in \mathbb{N} \longrightarrow \mathbb{N}$;

factorial(0) = 1 ;

$\forall n.(n \in \mathbb{N} \Rightarrow factorial(n + 1) = (n + 1) \times factorial(n))$

INITIALISATION

result : $\in \mathbb{N}$

ok := FALSE

EVENTS

computation = ANY *ok* = FALSE THEN *result, ok* := *factorial*(*m*), TRUE END

END

Modelling systems relations as events

MODEL

m

Static Part *context*

SETS

s

CONSTANTS

c

AXIOMS

$P(s, c)$

THEOREMS

$Q(s, c)$

Dynamic Part *machine*

VARIABLES

x

INVARIANT

$I(s, c, x)$

THEOREMS

$S(s, c, x)$

INITIALISATION

$Init(s, c, x)$

EVENTS

$\{r_1, \dots, r_n\}$

END

- $\Gamma(m)$ defines the static environment for the proofs related to m from s , c and $P(s, c)$ and $\Gamma(m)$ is defined from the static part.
- $\Gamma(m) \vdash Q(s, c)$
- $\Gamma(m) \vdash \forall x, x' \in \text{VALUES} : \text{INIT}(s, c, x) \Rightarrow I(s, c, x)$
- $\forall i \in \{1, \dots, n\} :$
 $\Gamma(m) \vdash \forall x, x' \in \text{VALUES} :$
 $I(s, c, x) \wedge x \ r_i \ x' \Rightarrow I(s, c, x')$
- $\Gamma(m) \vdash \forall x, x' \in \text{VALUES} : I(s, c, x) \Rightarrow S(s, c, x)$

Modelling systems relations as events

CONTEXT

context_name

SETS

s

CONSTANTS

c

AXIOMS

$P(s, c)$

THEOREMS

$Q(s, c)$

MACHINE

m

SEES

context_name

VARIABLES

x

INVARIANT

$I(s, c, x)$

THEOREMS

$S(s, c, x)$

INITIALISATION

$Init(s, c, x)$

EVENTS

$\{e_1, \dots, e_n\}$

END

- $\Gamma(m)$ defines the static environment for the proofs related to m from s , c and $P(s, c)$ and $\Gamma(m)$ is defined from the static part.
- $\Gamma(m) \vdash Q(s, c)$
- $\Gamma(m) \vdash \forall x, x' \in \text{VALUES} : \text{INIT}(s, c, x) \Rightarrow I(s, c, x)$
- $\forall i \in \{0, \dots, n\} : r_i(x, x') \triangleq BA(e_i)(s, c, x, x')$
- $\forall i \in \{0, \dots, n\} :$
 $\Gamma(m) \vdash \forall x, x' \in \text{VALUES} :$
 $I(x) \wedge BA(e_i)(s, c, x, x') \Rightarrow I(s, c, x')$
- $\Gamma(m) \vdash \forall x, x' \in \text{VALUES} : I(x) \Rightarrow S(s, c, x)$

step 1 : Understanding the **problem** to solve

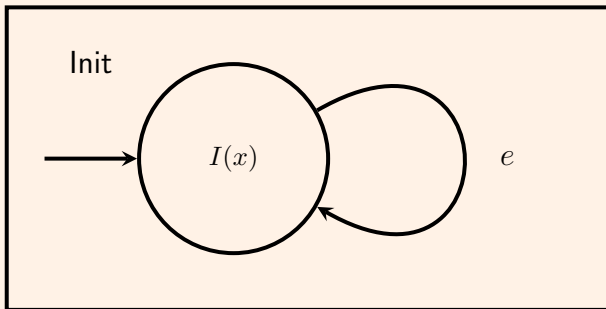
step 2 : Organizing requirements and extracting properties

step 3 : Writing a first very **abstract** system model

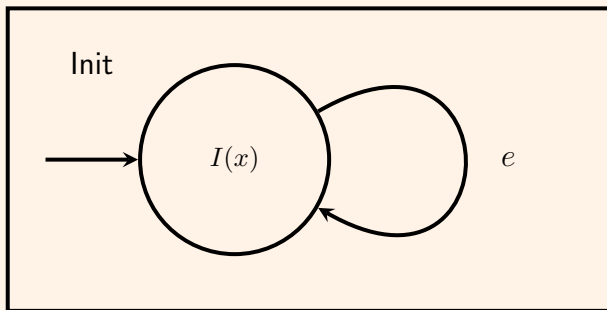
step 4 : Consulting the requirements and **adding** a new detail in the current model by **refinement**

step 5 : Either the model is enough detailed and the process stops, or the model is not yet enough concrete and the step 4 is repeated.

Summary on the internal checking



Summary on the internal checking



- I satisfies the two proof obligations
 - ▶ $Init(x) \Rightarrow I(x)$
 - ▶ $\forall e. I(x) \wedge BA(e)(x, x') \Rightarrow I(x')$
- A is a safety property satisfying $I(x) \Rightarrow A(x)$

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Expressing models in the Event B notation

- Models are defined in two ways :
 - ▶ an abstract machine
 - ▶ a refinement of an existing model
- Models use **constants** which are defined in structures called **contexts**
- B structures are related by the three possible relations :
 - ▶ the **sees** relationship for expressing the use of constants, sets satisfying axioms and theorems.
 - ▶ the **extends** relationship for expressing the extension of contexts by adding new constants and new sets
 - ▶ the **refines** relationship stating that a B model is refined by another one.

Machines

- **REFINES**
- **SEES** a context
- **VARIABLES** of the model
- **INVARIANTS** satisfied by the variables
- **THEOREMS** satisfied by the variables
- **EVENTS** modifying the variables
- **VARIANT**

Contexts

- **EXTENDS** another context
- **SETS** declares new sets
- **CONSTANTS** define a list of constants
- **AXIOMS** define the properties of constants and sets
- **THEOREMS** list the theorems which should be derived from axioms

MACHINE

m

REFINES

am

SEES

c

VARIABLES

u

INVARIANTS

I(u)

THEOREMS

Q(u)

VARIANT

< variant >

EVENTS

< event >

END

- $\Gamma(m)$: environment for the machine m defined by the context c
- $\Gamma(m) \vdash \forall u \in \text{VALUES} : \text{INIT}(u) \Rightarrow I(u)$
- For each event e in E :
 $\Gamma(m) \vdash \forall u, u' \in \text{VALUES} : I(u) \wedge BA(e)(u, u') \Rightarrow I(u')$
- $\Gamma(m) \vdash \forall u \in \text{VALUES} : I(u) \Rightarrow Q(u)$

CONTEXTS c **EXTENDS** ac **SETS** s **CONSTANTS** c **AXIOMS** $ax1 : \dots$ **THEOREMS** $th1 : \dots$ **END**

- $ac : c$ is extending ac and add new features
- s : sets are defined either by intension or by extension
- c : constants are defined and
- axioms characterize constants and sets
- theorems are derived from axioms in the current context

before-after relation for e

For each event e, a before-after relation is defined over (flexible) variables. Three events are possible

- $e \triangleq \text{BEGIN } x : |P(x, x') \text{ END} : \text{BA}(e)(x, x') \triangleq P(x; x')$
- $e \triangleq \text{WHEN } G(x) \text{ THEN } x : |P(x, x') \text{ END} :$
 $\text{BA}(e)(x, x') \triangleq G(x) \wedge P(x; x')$
- $e \triangleq \text{ANY } p \text{ WHEN } G(p, x) \text{ THEN } x : |P(p, x, x') \text{ END} :$
 $\text{BA}(e)(x, x') \triangleq \exists p. G(p, x) \wedge P(x; x')$

guard for e

For each event e, a guard is defined over (flexible) variables.

Three events are possible

- $e \triangleq \text{BEGIN } x : |P(x, x') \text{ END} : \text{grd}(x) \triangleq \text{TRUE}$
- $e \triangleq \text{WHEN } G(x) \text{ THEN } x : |P(x, x') \text{ END} : \text{grd}(e)(x) \triangleq G(x)$
- $e \triangleq \text{ANY } p \text{ WHEN } G(p, x) \text{ THEN } x : |P(p, x, x') \text{ END} : \text{grd}(e)(x) \triangleq \exists p. G(p, x)$

Proof obligations for a B model

$$\text{inv1} \quad \Gamma(s, c) \vdash \text{Init}(x) \Rightarrow I(x)$$

$$\text{inv2} \quad \Gamma(s, c) \vdash I(x) \wedge BA(e)(x, x') \Rightarrow I(x')$$

$$\text{fis} \quad \Gamma(s, c) \vdash I(x) \wedge \text{grd}(E) \Rightarrow \exists x' \cdot P(x, x')$$

safe $\Gamma(s, c) \vdash I(x) \Rightarrow A(x)$

$$\text{dead} \quad \Gamma(s, c) \vdash I(x) \Rightarrow (\text{grd}(e_1) \vee \dots \vee \text{grd}(e_n))$$

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Communicating agents

The factorial model

CONTEXT

fonctions

CONSTANTS

factorial, n

AXIOMS

ax1 : $n \in \mathbb{N}$

ax2 : $factorial \in \mathbb{N} \leftrightarrow \mathbb{N}$

ax3 : $0 \mapsto 1 \in factorial$

ax4 : $\forall(i, fn).(i \mapsto fn \in factorial \Rightarrow i + 1 \mapsto (i + 1) * fi \in factorial) \wedge$

$$\forall f \cdot \left(\begin{array}{l} f \in \mathbb{N} \leftrightarrow \mathbb{N} \wedge \\ 0 \mapsto 1 \in f \wedge \\ \forall(n, fn).(n \mapsto fn \in f \Rightarrow n + 1 \mapsto (n + 1) \times fn \in f) \\ \Rightarrow \\ factorial \subseteq f \end{array} \right)$$

END

The factorial model

MACHINE

specification

SEES *fonctions*

VARIABLES

resultat

INVARIANT

$resultat \in \mathbb{N}$

THEOREMS

$th1 : factorial \in \mathbb{N} \longrightarrow \mathbb{N} ;$

$th2 : factorial(0) = 1 ;$

$th3 : \forall n. (n \in \mathbb{N} \Rightarrow factorial(n+1) = (n+1) \times factorial(n))$

INITIALISATION

$resultat : \in \mathbb{N}$

EVENTS

$computing1 = \text{BEGIN } resultat := factorial(n) \text{ END}$

END

```

MACHINE factorialok SEES functions
VARIABLES r ok n
INVARIANTS
  @inv1 r ∈ ℤ
  @inv2 n ∈ ℤ
  @inv3 ok ∈ BOOL
  @inv4 n = n0
  @inv5 ok = TRUE ⇒ r = factorial(n) // postcondition
EVENTS
EVENT INITIALISATION
  then
    @act1 r : ∈ ℤ
    @act2 ok := FALSE
    @act3 n := n0
  end
EVENT computing1
  where
    @grd1 ok = FALSE
  then
    @act1 ok := TRUE
    @act2 r := factorial(n)
  end
end
end

```

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Communications between agents

MACHINE *agents*

SEES *data*

VARIABLES

sent

got

lost

INVARIANTS

$inv1 : sent \subseteq AGENTS \times AGENTS$

$inv2 : got \subseteq AGENTS \times AGENTS$

$inv4 : (got \cup lost) \subseteq sent$

$inv6 : lost \subseteq AGENTS \times AGENTS$

$inv7 : got \cap lost = \emptyset$

INITIALISATION

BEGIN

$act1 : sent := \emptyset$

$act2 : got := \emptyset$

$act4 : lost := \emptyset$

END

Communications between agents

EVENT sending a message

ANY

a, b

WHERE

$grd11 : a \in AGENTS$

$grd12 : b \in AGENTS$

$grd1 : a \mapsto b \notin sent$

THEN

$act11 : sent := sent \cup \{a \mapsto b\}$

END

EVENT getting a message

ANY

a, b

WHERE

$grd11 : a \in AGENTS$

$grd12 : b \in AGENTS$

$grd13 : a \mapsto b \in sent \setminus (got \cup lost)$

THEN

$act11 : got := got \cup \{a \mapsto b\}$

END

Communications between agents

```
EVENT loosing a message  
ANY  
  a  
  b  
WHERE grd1 : a ∈ AGENTS  
        grd2 : b ∈ AGENTS  
        grd3 : a ↦ b ∈ sent \ (got ∪ lost)  
THEN  
  act1 : lost := lost ∪ {a ↦ b}  
END
```

CONTEXTS

data

SETS

MESSAGES

AGENTS

DATA

CONSTANTS

n

infile

AXIOMS

axm1 : *n* ∈ ℕ

axm2 : *n* ≠ 0

axm3 : *infile* ∈ 1 .. *n* → *DATA*

END

Current Section Summary

General form of an event

```
EVENT e
  ANY t
  WHERE
     $G(c, s, t, x)$ 
  THEN
     $x : |(P(c, s, t, x, x'))|$ 
  END
```

- c et s are constantes and visible sets by e
- x is a state variable or a list of variables
- $G(c, s, t, x)$ is the condition for observing e .
- $P(c, s, t, x, x')$ is the assertion for the relation over x and x' .
- $BA(e)(c, s, x, x')$ is the *before-after* relationship for e and is defined by $\exists t. G(c, s, t, x) \wedge P(c, s, t, x, x')$.

General form of proof obligations for an event e

Proofs obligations are simplified when they are generated by the module called POG and goals in sequents as $\Gamma \vdash G$:

- ① $\Gamma \vdash G_1 \wedge G_2$ is decomposed into the two sequents

$(1) \Gamma \vdash G_1$
 $(2) \Gamma \vdash G_2$
- ② $\Gamma \vdash G_1 \Rightarrow G_2$ is transformed into the sequent $\Gamma, G_1 \vdash G_2$

Proof obligations in Rodin

- $INIT/I/INV : C(s, c), INIT(c, s, x) \vdash I(c, s, x)$
- $e/I/INV : C(s, c), I(c, s, x), G(c, s, t, x), P(c, s, t, x, x') \vdash I(c, s, x')$
- $e/act/FIS : C(s, c), I(c, s, x), G(c, s, t, x) \vdash \exists x'. P(c, s, t, x, x')$

- Chapter Event B
- The Event B Modelling Notation Version 1.4
- The Event-B Mathematical Language 2006
- User Manual of the RODIN PPlatform